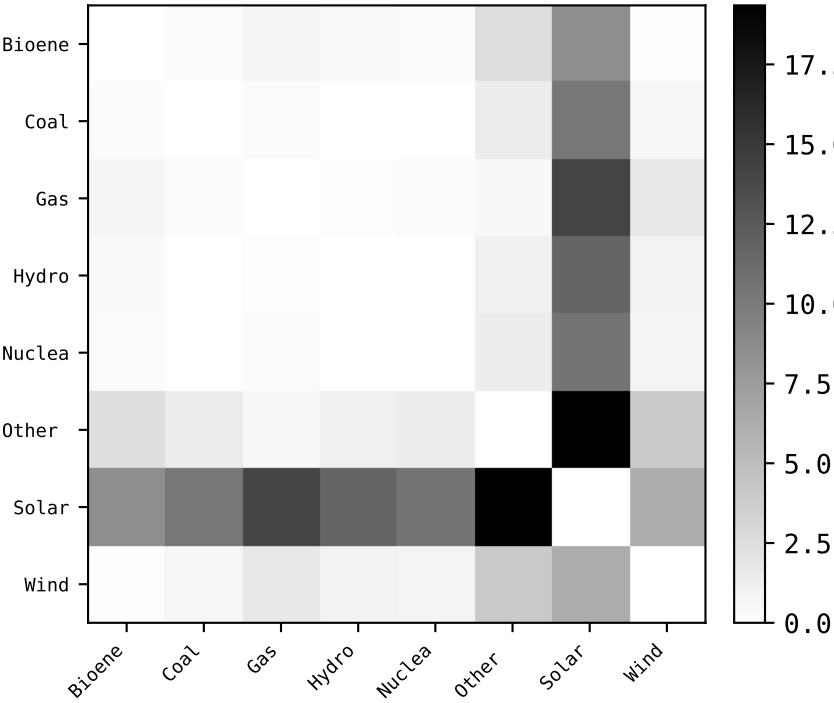


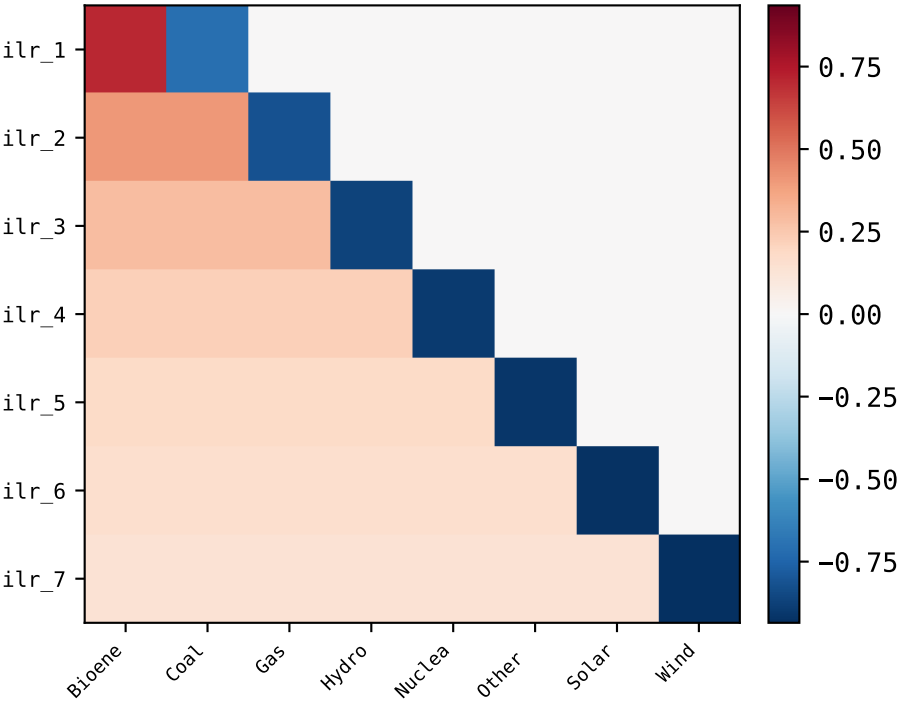
Foundations Plate (Stage 0) – EMB IND India generation TWh.csv · D = 8 carriers · N = 26 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible

§1 Variation matrix $\text{var}(\log x_i/x_j)$
symmetric: $\max|M - M^T| = 0.00\text{e}+00$



§2 Helmert basis H (D-1 × D)
orthonormality: $\max|HH^T - I| = 2.22\text{e}-16$

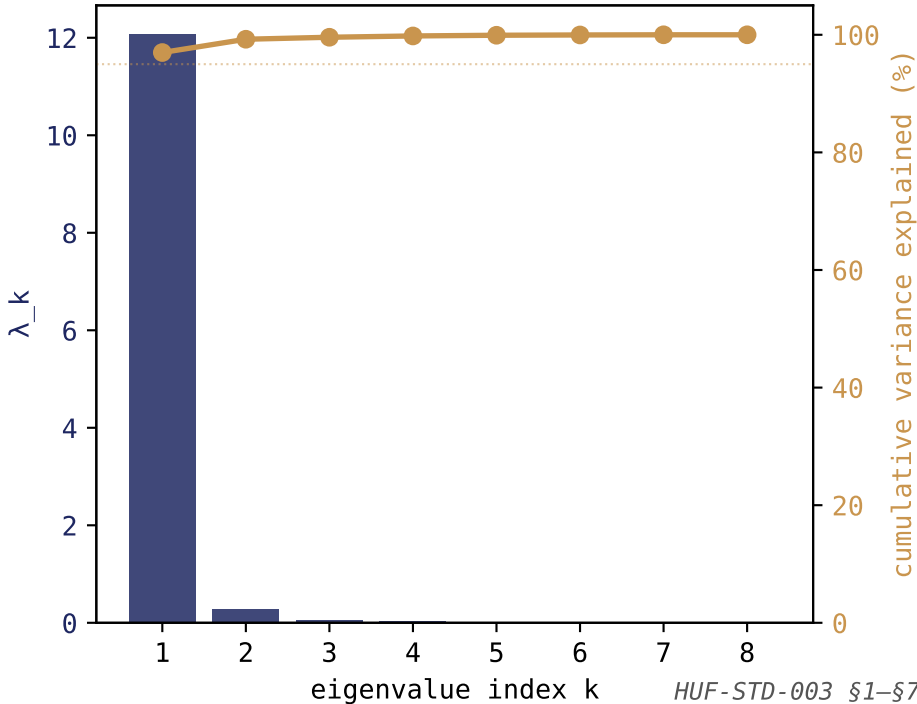


§3 Matrix Decomposition Chain

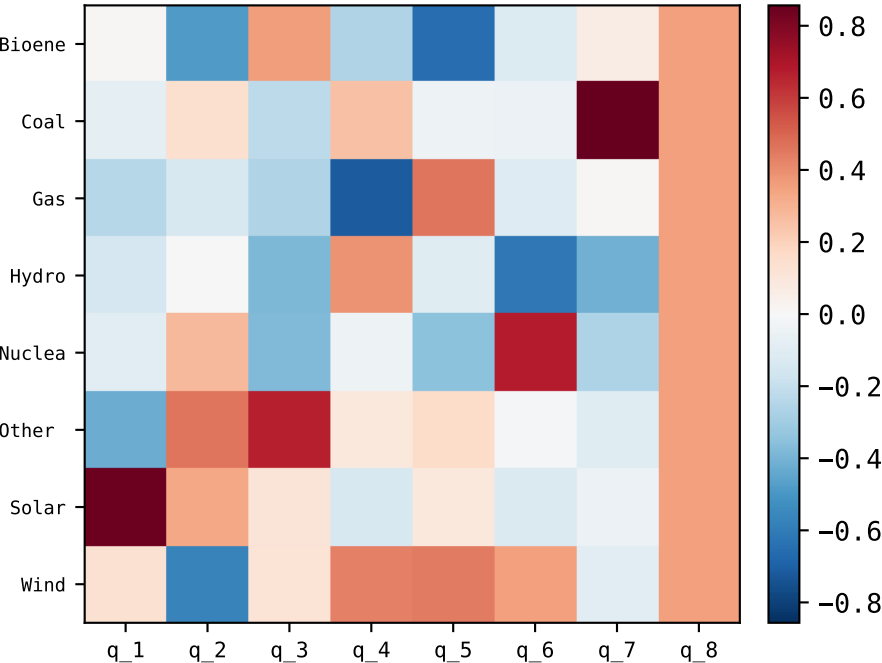
raw composition
↓ closure $C(x) = x / \Sigma x$
closed composition $x \in \Delta^{D-1}$
↓ CLR: $x \rightarrow \log x - \langle \log x \rangle$
CLR coordinates (R^8)
↓ ILR: $\text{clr} \rightarrow \text{clr} @ H^T$
ILR coordinates (R^7)

- D = 8 carriers
- N = 26 timesteps
- each arrow is exact (modulo IEEE float)

§4 Eigenvalue scree (CLR covariance)
top $\lambda = 12.061$ · trace = 12.436



§6 Orthonormal eigenbasis Q
columns = eigenvectors, ordered by $\lambda \downarrow$



§5 Spectral Theorem verification

$\Sigma = Q \Lambda Q^T$ (theorem)
 $\max|\Sigma - Q \Lambda Q^T| = 1.07\text{e}-14$
(IEEE-floor $\rightarrow$ theorem holds)

 Σ symmetric?
 $\max|\Sigma - \Sigma^T| = 0.00\text{e}+00$

All λ real?
 $\max|\text{imag}(\lambda)| = 0.00\text{e}+00$ (eigh $\Rightarrow$ real)

Q orthonormal?
 $\max|Q^T Q - I| = 8.88\text{e}-16$

Rank-k variance explained:
k=1 $\rightarrow$ 97.0%
k=2 $\rightarrow$ 99.2%
k=3 $\rightarrow$ 99.6%
k=8 $\rightarrow$ 100.0%

Foundations – Numeric Verification – EMB IND India generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

FOUNDATION	STATEMENT	VALUE	JUDGMENT
\$1 Symmetric	$\max M - M^T $ for variation matrix	0.000e+00	exact (= 0)
\$1 Symmetric	$\max \Sigma - \Sigma^T $ for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
\$2 Transpose	$\max H H^T - I_{\{D-1\}} $ (Helmert orthonormal rows)	2.220e-16	IEEE-floor
\$2 Transpose	$\max H^T H - P_{\{D-1\}} $ (projection onto image)	2.220e-16	IEEE-floor
\$3 Decomp	closure $\rightarrow$ CLR $\rightarrow$ ILR $\rightarrow$ CLR roundtrip residual	3.750e-09	IEEE-floor
\$4 Eigen	number of eigenvalues found	8	= D (8)
\$4 Eigen	$\max \text{imag}(\lambda_k) $ (real because Σ symmetric)	0.000e+00	exact
\$4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	12.0611	–
\$4 Eigen	$\text{trace}(\Sigma) = \sum \lambda_k$ (total variance)	12.4363	–
\$5 Spectral	$\max \Sigma - Q \Lambda Q^T $ (Spectral Theorem residual)	1.066e-14	IEEE-floor
\$5 Spectral	$\max Q^T Q - I $ (orthonormality of eigenbasis)	8.882e-16	IEEE-floor
\$6 Decomp	cumulative variance at k=1 (rank-1 approximation)	96.98%	–
\$6 Decomp	cumulative variance at k=2	99.24%	–
\$6 Decomp	cumulative variance at k=3	99.58%	–
\$7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

The foundations carry the bedrock. The instrument reads. The expert decides.
The hashes carry the receipts. The vocabulary holds the line.